PROPOSED SCORING GUIDE AND RUBRIC

UMTA UACE - 2026 PAPER 1 - CHEMISTRY.

TEL: 0777376396/0752672838

"The chemistry Medicine".

TR. OPELE DANIEL

Signature.

PAGE NO:

ITEM 1.

Time (minutes)	0	10	20	30	40	50	60
Concentration (mole/m^3) of dichromate (VI) ions	2.500	1.581	1.090	0.631	0.420	0.25	0.160
log (concentration of dichromate (VI) ions)	0.3979 ✓	0.1989 ✓	0.0374 ✓	-0.2000 ✓	-0.3768 ✓	-0.6021 ✓	-0.795 ✓

✓Da - Finding all log (concentration of dichromate (VI) ions)

From graph, slope = -0.020105

$$\text{slope} = \frac{-k}{2.303} \quad \checkmark Di$$

$$k = 2.303 \times 0.020105$$

$$k = 0.046302 \text{ minute}^{-1} \quad \checkmark Di$$

$$t_{\frac{1}{2}} = \frac{0.693}{k} \quad \checkmark Di$$

$$t_{\frac{1}{2}} = 14.96696 \text{ minutes}$$

$$\simeq 15 \text{ minutes.} \quad \checkmark Di_{output}$$

Rej; with no incorrect units.

The half life of dichromate (VI) ions is 15 minutes; it takes 15 minutes for dichromate (VI) ions to reduce to half its original concentration. ✓Dc This is more than 10 minutes implying that dichromate (VI) ions is more efficient to be used as a disinfectant. ✓Dc

7Da + 6Di + 2Dc = 03
7Da + 6Di = 02
7Da + 1-5Di = 02
7Da = 01

Da - Data analysis
Di - Data interpretation
Dc - Drawing conclusion

Score = 00 when Di_{output} is incorrect. Consider any other correct alternative

PICFARE

The solution stored in outreach box during 'hot afternoon' is in hot conditions with higher temperature ✓Fe therefore particles in solution have higher kinetic energy, ✓Fe causing more frequent and energetic collisions ✓Fe that speed up the reduction of dichromate (VI) ions, $Cr_2O_7^{2-}$ to chromium (III) ions. ✓Fe This reduces the concentration of the active oxidising agent ($Cr_2O_7^{2-}$) making it ineffective sooner. ✓Fe

Mean while refrigeration lowers particles kinetic energy reducing ✓Fe collision frequency and slowing the reduction process so the solution was effective for longer time. ✓Fe

✓L - Collision theory used.

Hydration is an exothermic reaction. The higher the hydration energy, the more negative the electrode potential. ✓Fe, This is because more energy is given off when gaseous ions are completely surrounded by water molecules. ✓Fe, The change in electrode potentials alters cell emf and affect the efficiency/operation of the sterilizer. ✓Fe,

The more positive the emf of a cell, the faster/better the sterilization process ✓Fe, and the less positive the emf of a cell, the lower/slower the sterilization process. ✓Fe,

✓L- Electrochemical theory / "ion in solution become surrounded by water molecules/etc ____."

✓L - Thermodynamic principle - "More energy is given off when ____"

3L + 7-8Fe + 5Fe, = 03
3L + 5-6Fe + 3-4Fe, = 02
3L + 1-4Fe + 1-2Fe, = 01

"No 3L = Score 00"
<3L
< 2 factors = 00

L- Theory
F - Factors affecting chemical

PAGE NO:

$Pt_{(s)} / Cl^-_{(aq)} / Cl_{2(g)} // Cr_2O_7^{2-}{}_{(aq)}, H^+_{(aq)}, Cr^{3+}_{(aq)} / Pt_{(s)}$ ✓Ea

$E^\theta_{cell} = E^\theta_{cathode} - E^\theta_{anode}$. ✓Ea

Acc $E^\theta_{cell} = E^\theta_{RHS} - E^\theta_{LHS}$.

$E^\theta_{cell} = (+1.53) - (+1.36)$

$E^\theta_{cell} = +0.17\ V$ ✓Ea Rej +0.17V, +0.17V without units.

Free energy change.

$\Delta G^\theta = -nFE^\theta_{cell}$. ✓Ea

$Cr_2O_7^{2-}{}_{(aq)} + 14H^+_{(aq)} + 6Cl^-_{(aq)} \rightarrow 2Cr^{3+}_{(aq)} + 3Cl_{2(g)} + 7H_2O_{(l)}$

$n = 6$

$\Delta G^\theta = -6 \times 96500 \times (+0.17)$ ✓Ea

$\Delta G^\theta = -98430$ Joules ✓Ea

OR $= -98.43kJ$.

The free energy change is $-98.43kJmol^{-1}$. The negative value shows that the cell reaction is feasible, ✓EC so the power unit can operate without an external energy source.

In hot conditions, ✓EC the concentration of dichromate (VI) ions decreases, ✓EC the reduction potential of the dichromate electrode becomes less positive; ✓EC the cell emf decreases, ✓EC the free energy change becomes less negative, ✓EC the sterilizer produces less electrical energy, ✓EC the disinfectant also loses ✓EC oxidizing power; Hence the disinfectant should be stored in a cool place ✓EC where decomposition is slower and the dichromate (VI) ions concentration remains higher

PAGE NO:

Platinum electrodes should be used because the reacting species are ions and gases with no conducting solid present; Platinum is chemically inert, conducts electricity well and does not react with dichromate (VI) ions or chlorine giving accurate and stable electrode potentials.

6Ea + 12Ec = 03
6Ea + 9-12Ec) = 02
6Ea + 1 - 8Ec) = 01
6Ea

Ea - Applying concept
Ec - Predicting chemical behaviour / optimizing industrial conditions and recomendations.

Item No

UGANDA NATIONAL EXAMINATIONS BOARD

(To be fastened together with other answers to paper)

UCE

Candidate s Name .....

Signature .....ITEM 1.....

Subject ..... Paper code ...../...

Random No.					
Personal No.					

✓0a

A graph of log (concentration of dichromate (VI) ions) against time

Scale

Horizontal axis

1cm represents 5 minutes

Vertical axis

1cm represents 50×10^{-3}

Shape – ✓1a

All plots correct – ✓1a

Scale – ✓1a

log (concentration of dichromate (VI) ions) $\times 10^{-3}$

0, -100, -200, -300, -400, -500, -600, -700

0, 10, 20, 30, 40, 50, 60, 70, 80

Time (minutes) ✓1a

✓1a

$(30, -0.200)$ x_1, y_1

$(50, -0.6021)$ x_2, y_2

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope} = \frac{-0.6021 - (-0.2)}{50 - 30}$$ ✓1a

$$\text{Slope} = \frac{-0.4021}{20}$$

$$= -0.020105$$ ✓1a

ITEM 2 3T+1Te+1Fe, =01

Direct heating lowers eucalyptus oil yeild by causing excessive evaporation ✓Te, loss and decomparition ✓Te of volatile oil components and reduces aroma by breaking down ✓Te the aromatic compaunds responsible for its smell ✓T. Laws- Kinetic molecular theory ✓T, vapour pressure principle ✓T, thermal decomparition principle ✓T

No buffer causes fluctuation in acidity and alter pH of the antiseptic; affecting stability and effectiveness of the active ingredients ✓Te. This reduce anti-microbial activity ✓Te, cause chemical degradation and shorten the shelf-life of the antiseptic (final product). ✓Te, ✓T

Laws - Acid-base buffer theory ✓T, chemical equilibrium ✓T, stability principle ✓T

Temperature (°C)	70	80	90	100	110
Vapour pressure of Q (kPa)	1.4	2.4	3.9	6.1	9.2
Vapour pressure of water (kPa)	19.0	47.0	72.0	104.0	132.0
Total vapour Pressure of mixture (kPa)	20.4 ✓	49.4 ✓	75.9 ✓	110.1 ✓	141.2 ✓

✓Ca - All Total vapour pressure values with units.

From graph; P_Q (Vapour pressure of Q at distillation temperature)

$P_Q = 5.50$ kpa ✓Ci

P_{H_2O} (Vapour pressure of water at distillation temperature)

$P_{H_2O} = 95$ kpa ✓Ci

Molar mass of $H_2O = (2\times1) + (16\times1)$

$= 18$ g

Percentage comparition of water in the distillate $= (100 - 3.43)$

$= 96.57\%$ ✓Ci

3T + 3Tet 3Fe, = D3
3T + 2Tet 2FeTe, = D2
3T + Tet Fe Te, = D1

ITEM 2

Direct heating lower eucalyptus oil yield by causing excessive evaporation, loss and decomposition of volatile oil components and reduces aroma by breaking down the aromatic compounds responsible for its smell. Le, Louis Kinetic molecular theory, vapour pressure principle, thermal decomposition principle

No buffer causes fluctuation in acidity and alter pH of the antiseptic affecting stability and effectiveness of the active ingredients. This reduce antimicrobial activity, cause chemical degradation and shorten the shelf-life of the antiseptic (final product). Le, Louis - Acid-base buffer theory, chemical equilibrium, stability principle

Temperature (°C)	70	80	90	100	110
Vapour pressure of O (kPa)	1·4	9·4	3·9	8·1	9·2
Vapour pressure of water (kPa)	19·0	47·0	72·0	104·0	132·0
Total vapour Pressure of mixture (kPa)	20·4	49·4	75·9	110·1	141·2

Ca - All Total vapour pressure values with units.

From graph: P_O (Vapour pressure of O at distillation temperature)

$P_O = 55.0$ kPa Ci

P_{H_2O} (Vapour pressure of water at distillation temperature)

$P_{H_2O} = 95$ kPa Ci

Molar mass of $H_2O = (2 \times 1) + (16 \times 1)$

$= 18$ g

Percentage composition of water in the distillate $= (100 - 3.4)\%$

$= 96.6\%$ Ci

3.43% CORRECTED TO 23.0%

$$\frac{P_{H_2O}}{P_Q} = \frac{n_{H_2O}}{n_Q}$$ ✓ci

$$\frac{95}{5.5} = \frac{77 \cdot M_r Q}{18 \times 23}$$ ✓ci

$$M_r Q = 92.8689$$
$$\simeq 939$$

∴ The molecular mass of Q is 939. ✓ci

Mass of sample Q = 0.56g

Mass of carbon = $\frac{12}{44}$ of 1.591
= 0.43399. ✓ca

Mass of Hydrogen = $\frac{2}{18}$ of 0.379
= 0.042111g. ✓ca

Mass of nitrogen = 0.56 − (0.43397 + 0.042111)
= 0.083999. ✓ca

Elements present;	C	H	N
Mass composition;	0.4339	0.042111	0.08399
Moles;	$\frac{0.4339}{12}$	$\frac{0.042111}{1}$	$\frac{0.08399}{14}$
	0.03616	0.042111	0.005999
Simplest mole ratio.	$\frac{0.03616}{0.005999}$	$\frac{0.042111}{0.005999}$	$\frac{0.005999}{0.005999}$
	6.027 ≃ 6	7.0197 ≃ 7	1 ✓ca

Empirical formula is C_6H_7N ✓ca

Molecular formula is $(C_6H_7N)_n$ = Molecular weight ✓ci

$(12 \times 6 \times n) + (1 \times 7 \times n) + (14 \times 1 \times n) = 93$

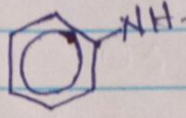
$93n = 93$

$n = 1$ ✓ ci

Molecular formula is the same as empirical formula

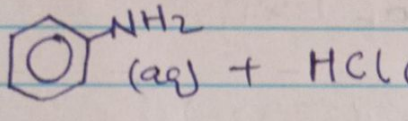
Molecular formula of Q C_6H_7N ✓ ci

Therefore

Since Q is made up of 6 carbon atoms, 7 hydrogen atoms and 1 nitrogen atom, therefore the structural formula of Q is  ✓ ci OR $C_6H_5NH_2$

which is Phenylamine.

PH of blending solution

 ✓ Ca

Moles of hydrochloric acid that reacted $= \left(\frac{0.1 \times 35}{1000}\right)$ moles ✓ Ca

$= 0.0035$ moles

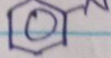
Moles of $C_6H_5NH_2$ that reacted $= \left(\frac{0.1 \times 45}{1000}\right)$ moles

$= 0.0045$ moles ✓ Ca

Moles of excess $C_6H_5NH_2$

$= (0.0045 - 0.0035)$

$= 0.001$ moles. ✓ Ca

Moles of Phenylammonium chloride,  formed

$= 0.0035$ moles ✓ Ca

Total volume $(35 + 45) = 80 cm^3$.

$80 cm^3$ of solution contains 0.001 moles of $C_6H_5NH_2$

$1000 cm^3$ of solution contains $\left(\frac{0.001 \times 1000}{80}\right)$ moles of $C_6H_5NH_2$

$= 0.0125 M$ ✓ Ca

$80cm^3$ of solution contains 0.0035 moles of $C_6H_5NH_3Cl$

$1000cm^3$ of solution contains $\left(\frac{0.0035 \times 1000}{80}\right)$ moles of $C_6H_5NH_3Cl$

$= 0.04375M$. ✓ca

from ∴ $pH = 14 - \left(pKb + \log \frac{[salt]}{[base]}\right)$ ✓ci

$$pH = 14 - \left[-\log(4.2 \times 10^{-10}) + \log\left(\frac{0.04375}{0.0125}\right)\right]$$ ✓ci

$$pH = 14 - (9.3768 + 0.5441)$$

$$pH = 4.0791$$

$$\simeq 4.1$$ ✓ci

The pH of the blending solution is 4.1. The blending solution contains a mixture of $C_6H_5NH_2$ (Phenylamine) and $C_6H_5NH_3Cl$ (Phenylammonium chloride) which forms a buffer solution ✓cc with a pH of 4.1.

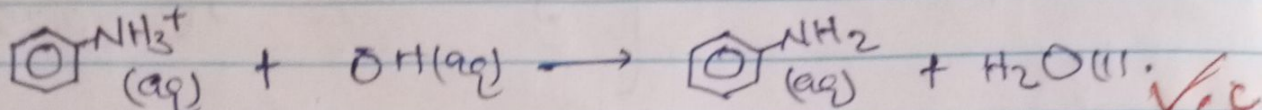
When a little sodium hydroxide solution accidentally poured into a blending solution before packaging, the solution resist ✓cc an increase in PH because phenylammonium chloride in solution is a strong ✓ci electrolyte hence fully ionizes to form phenylammonium ions and chloride ions; ✓cc Phenylamine is a weak base hence only slightly ionizes ✓cc and its ionisation is suppressed by the high ✓cc concentration of phenylammonium ions from the salt

$$C_6H_5NH_3Cl_{(aq)} \longrightarrow C_6H_5NH_3^+{}_{(aq)} + Cl^-_{(aq)}$$ ✓ci

$$C_6H_5NH_{2(aq)} + H_2O_{(l)} \rightleftharpoons C_6H_5NH_3^+$$ ✓ce

The solution therefore contains a high concentration of phenylammonium ions and a high concentration of unionised phenylamine molecules ✓Cc

When little sodium hydroxide is added, the hydroxide ions reacts with phenylammonium ions to form phenylamine molecules and water, ✓Cc resisting an increase in pH. ✓Cc This make the technician guided that the little addition should not worry them as the solution's pH will not be affected. ✓Cc



19 Ca + 14 Ci + 12 Cc = 03
19 Ca + 14 Ci } = 02
19 Ca + 1-13 Ci }
19 Ca = 01.

Ca – Data analysis
Ci – Data interpretation
Cc – Explanation of system behaviour.

Buffers such as a mixture of phenylammonium chloride and phenylamine preserve final ✓Ai antiseptic product by maintaining constant pH, however the buffer may irritate the skin and eyes causing redness, pain and inflamation to factory workers. ✓Ar

Phenylamine used is toxic and can be absorbed through the skin causing skin and eye irritation ✓Ar especially with repeated exposure however it is used to synthesize several drugs and pharmaceutical intermediates. ✓Ai

2 Ai + 2 Ar = 02
2 Ai = 01

application:
Ai – Use of chemical system.
Ar – Danger/side effect of chemical system

Item No

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(To be fastened together with other answers to paper)

UCE

Candidate s Name .....

Signature .....

Subject ..... Paper code...../....

Random No.

Personal No.

A graph of vapour pressure against temperature.

Scale
Horizontal axis
1 cm² represents 2.5°C
Vertical axis
1cm represents 10kPa

$P_Q = 5.5kPa$

$P_{H_2O} = 95kPa$

Sharper and named

All plots correct

Scale

Axes with units

Total Vapour pressure of mixture

Vapour pressure of water

Vapour pressure of Q

Vapour Pressure (kPa)

Temperature (°C)

180
160
140
120
100
80
60
40
20
0

70
75
80
85
90
95
100
105
110

SECTION B

Part I

ITEM 3

The tendency to become less positive in the standard electrode potential values increases from fluorine to chlorine to iodine and the oxidising strength decreases from fluorine to iodine; This is because ionic radius increases from fluorine to iodine which decreases the standard electrode potential of the ions. Therefore fluoride and chloride ions have smaller radius and more positive standard electrode potential values hence fluoride and chloride ions are regarded as stronger oxidizing agents and the strength is in the order $F_2 > Cl_2 > I_2$. Fluorine is the most oxidising ingredient making it the best disinfectant or surface-cleaning agent basing on this trend.

The boiling point increases in the order fluorine < chlorine < iodine; This is because atomic radius increases, which increases the molecular weight and hence the strength of Vander Waal's forces of attraction increases in the order $F_2 < Cl_2 <$ [illegible] I_2. Therefore the energy required to overcome these forces increases from fluorine to iodine. Iodine with the highest boiling point means it is less volatile so it is retained more easily during storage and use.

Being oxidizing agent, fluorine reacts vigorously

13Pe = 02 = 8Pe + 7Pe,

Pe – Well explained physical property.

Rej: Re - Unexplained reactions
Eq - Unbalanced equation.

Being oxidizing agent, fluorine reacts vigorously with water, oxidizing it to oxygen and itself reduced to hydrofluoric acid ✓Re

$$2F_2(g) + 2H_2O(l) \longrightarrow 4HF(aq) + O_2(g).$$ ✓Eq

Chlorine reacts slowly and reversibly in a disproportionation reaction where chlorine is oxidized and reduced in the same reaction to form hydrochloric acid and hypochlorous acid. ✓Re

$$Cl_2(g) + H_2O(l) \rightleftharpoons HCl(aq) + HOCl(aq)$$ ✓Eq

OR

$$Cl_2(g) + H_2O(l) \rightleftharpoons 2H^+(aq) + Cl^-(aq) + ClO^-(aq).$$

When the resultant solution is exposed to sunlight, the hypochlorous acid decomposes with liberation of oxygen gas ✓Re

$$2HOCl(aq) \longrightarrow O_2(g) + 2HCl(aq).$$ ✓Eq

Iodine, however, is just sparingly soluble in water and does not react with water ✓Re

Fluorine reacts with cold dilute sodium hydroxide by oxidizing hydroxide ions to oxygen and itself reduced to fluoride ions. Water is also given off during the reaction. ✓Re

$$2F_2(g) + 2\bar{O}H(aq) \longrightarrow OF_2(g) + 2F^-(aq) + H_2O(l).$$ ✓Eq

OR

$$2F_2(g) + 2NaOH(aq) \longrightarrow OF_2(g) + 2NaF(aq) + H_2O(l).$$

Chlorine and iodine reacts with cold dilute sodium hydroxide in a disproportionation reactions where they are reduced and oxidized in the same reaction to form halide and halate(I)

✓Re ✓Eq

$Cl_2(g) + 2\bar{O}H(aq) \longrightarrow Cl^-(aq) + ClO^-(aq) + H_2O(l)$.

✓Eq

$I_2(s) + 2\bar{O}H(aq) \longrightarrow Br^-(aq) + BrO^-(aq) + H_2O(l)$.

Although fluorine has the highest standard electrode potential making it the strongest ✓RI oxidising agent, it reacts violently with water and sodium hydroxide, is highly toxic ✓RI and has a very low boiling point making it unsafe and difficult to handle. ✓RI On the other hand, iodine has the highest ✓RI boiling point making it easiest to store and handle safely, but it's low standard electrode potential ✓RI make it the weakest oxidising agent and therefore a less effective disinfectant. ✓RI

Chlorine is the most suitable active ingredient because it has a sufficiently high standard electrode potential, reacts with water and dilute sodium hydroxide to form hypochlorite ✓RI ions, the active disinfecting species and is safer to use than fluorine while being more effective than iodine. ✓RI

6Re + 6Eq + 8RI = 03

Chlorine

6Re + 6Eq = 02

Re - Explained chemical reactions
Eq - Balanced chemical equation.
RI - Interpretation.

- Reason for the reaction
- Condition
- Type of reaction
- Products name, etc.

- Correct chemical symbols
- Correct states
- Correct balancing

Chlorine as a disinfectant has advantage that it reacts with water to form an acidic solution which kills germs making surface clean once applied however if use is excess and disposed to water bodies, it lowers the PH of water significantly causing death of aquatic organisms. This can be mitigated by ensuring proper disposal and use of controlled amounts of chlorine.

Ea Ed EM

Chlorine is used to make bleaching compounds used for whitening paper, textiles, etc however chlorinated wastewater can alter microbial communities in soil and water, affecting natural nutrient cycles. This can be mitigated by dechlorination of wastewater i.e treating chlorinated waste water before discharge.

Ea Ed EM

2Ea+2Ed+2EM = 03
2Ea+2Ed = 02
2Ea/2Ed = 01

Ea - Explained application
Ed - Explained danger
EM - Explained Mitigation

ITEM 4.

Calcium is the most lightweight metal; Density decreases from beryllium to calcium then increases from calcium to barium. ✓Pe

From beryllium to calcium; as atomic number increases, atomic radius increases. ✓Pe The increase in atomic volume is greater than the increase in atomic mass ✓Pe and so the atoms are packed less closely hence density decreases. ✓Pe

From calcium to barium; although atomic size continues to increase ✓Pe, the atomic mass increases much more rapidly than the atomic volume. ✓Pe Therefore, more mass is contained in each unit volume ✓Pe, causing density to increase; making calcium the lightest. ✓Pe

The melting point generally increases from scandium to chromium; ✓Pe then decreases from manganese to copper; From scandium to vanadium, the number of unpaired electron(s) in the 3d-sub energy level increases, ✓Pe strengthening metallic bonding because more delocalized electrons participate in bonding. ✓Pe, Stronger metallic bonds require more energy to break, so the melting point rises. ✓Pe, Chromium melting point is slightly lower than that of vanadium due to differences in crystal structure and overall nature of metallic bonding. ✓Pe,

Melting point decreases from manganese to copper. Manganese have abnormally low melting point because manganese has a half-filled 3d sub-energy level ✓Pe, thermodynamically stable making the 3d-electrons less available for metallic bonding. ✓Pe, It only uses 4s electrons per atom in metallic bonding.

Manganese – $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$.

Melting point decreases because the number of unpaired electrons in the 3d-sub energy level decreases, ✓Pe, weakening the metallic bonds. ✓Pe,

8Pe + 9Pe, = 02

TRANSITION METALS.

Reactions with water.

Chromium being more reactive that hydrogen removes oxygen from steam to form chromium(III) oxide while hydrogen gas is liberated. ✓Re

$2Cr(s) + 3H_2O(g) \longrightarrow Cr_2O_3(s) + 3H_2(g)$. ✓Eq

Manganese reacts with hot water in a single-displacement redox reaction where manganese is oxidized as it transfers electrons to water displacing hydrogen gas and forming manganese(II) hydroxide. ✓Re

$Mn(s) + H_2O(l) \longrightarrow Mn(OH)_2(s) + H_2(g)$. ✓Eq

Strongly heated titanium reacts with steam in a single-displacement redox reaction to form titanium(IV) oxide and hydrogen gas ✓Re

$Ti(s) + 2H_2O(g) \longrightarrow TiO_2(s) + 2H_2(g)$. ✓Eq

When red-hot cobalt or nickel reacts with steam in a single-displacement redox reaction, forms their respective oxides liberating hydrogen gas. ✓Re

$Co(s) + H_2O(g) \longrightarrow CoO(s) + H_2(g)$. ✓Eq

$Ni(s) + H_2O(g) \longrightarrow NiO(s) + H_2(g)$. ✓Eq

Scandium, Vanadium and copper does not react with water.

Chromium reacts with hot sodium hydroxide solution to form a chromite and hydrogen gas in a single displacement redox reaction. ✓Re

$2Cr(s) + 6\bar{O}H(aq) \longrightarrow 2CrO_3^{3-}(aq) + 3H_2(g)$. ✓Eq

The rest does not react with sodium hydroxide solu-

GROUP 2 ⇒ II

Beryllium doesn't react with water under any condition due to the insolubility of beryllium hydroxide and formation of protective oxide layer; impermeable and prevents water from reaching the underlying metal. ✓Re

Magnesium reacts slowly with cold water; because it is covered by a thin layer of magnesium oxide which protects the metal from further attack. This reaction form Magnesium hydroxide and hydrogen gas ✓Re

$$Mg(s) + 2H_2O(l) \longrightarrow Mg(OH)_2(aq) + H_2(g)$$ ✓Eq

When heated, magnesium reacts with steam because the high temperature provides enough energy to overcome the protective oxide layer and allows magnesium to lose electrons readily, reducing water to hydrogen gas and forming magnesium oxide. ✓Re

$$Mg(s) + H_2O(g) \longrightarrow MgO(s) + H_2(g)$$ ✓Eq

Calcium reacts Moderately with cold water because it loses its two outer electrons easily due to its larger atomic size and lower ionization energy forming calcium hydroxide and releasing hydrogen gas ✓Re

$$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$$ ✓Eq

Barium reacts vigorously with cold water because it has a large atomic radius and low ionization energy, so it loses its two outer electrons very easily, forming soluble barium hydroxide and releasing hydrogen gas. ✓Re

$$Ba(s) + 2H_2O(l) \longrightarrow Ba(OH)_2(aq) + H_2(g)$$ ✓Eq

Beryllium reacts with hot concentrated sodium hydroxide solution because it is amphoteric; it reacts with strong alkalis to form sodium beryllate and releases hydrogen gas. ✓Re

$$Be(s) + 2NaOH(aq) + 2H_2O(l) \longrightarrow Na_2Be(OH)_4(aq) + H_2(g)$$ ✓Eq

OR

$$Be(s) + 2\bar{O}H(aq) + 2H_2O(l) \longrightarrow Be(OH)_4^{2-}(aq) + H_2(g).$$

OR

$$Be(s) + 2\bar{O}H(aq) \longrightarrow BeO_2^{2-}(aq) + H_2(g).$$

Structurally and chemically beryllium would be better than magnesium because beryllium because it is lightweight, recoverable with alkalis and does not react with water ✓Rf1, however beryllium is toxic and harzardous to the environment so magnesium is a better material because it is lightweight, non-toxic and safe for school laboratory setting. ✓Rf1

Vanadium would be better than chromium because vanadium have the highest melting point compared to other transition metals ✓Rf2 however chromium's exceptionally high melting point provides nearly identical thermal resistance with added advantage of forming a stable protective surface oxide layer that prevents high temperature oxidation making it most durable and reliable material under practical laboratory heating conditions. ✓Rf2.

11Re + 11Eq + 2Rf1 + 2Rf2 = 03

Chromium is used in the manufacture of stainless steel because of its self healing layer of chromium oxide passive layer which prevents the iron in the steel from rusting. ✓Ea However during industrial smelting, chromium(III) can oxidize into hexavalent chromium which is highly soluble allowing it to easily leach through soil and contaminate groundwater tables and surface rivers ✓Ed leading to death of aquatic animals since it is toxic. This can be mitigated by ensuring proper recycling of worn-out chromium vessels. ✓Em

Beryllium is used to manufacture aerospace components like mirrors on telescope ✓Ea because it is lightweight however it is highly toxic when inhaled causing chronic fatal lung disease. ✓Ed This can be mitigated by using an alternative safe material like magnesium. ✓Em

2Ea + 2Ed + 2Em = 03

2Ea + 2Ed = 02

2Ea | 2Ed = 01

UMTA SCORING GUIDE AND RUBRIC (PROPOSED)

Part II

Item 5.

(Intermediate)

Explanation why linear alkylbenzene is suitable for the manufacture of the biodegradable household detergent

The linear alkyl chain is easily broken down by microorganisms, making the resulting detergent biodegradable. ✓Pu

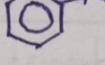
The long hydrocarbon chain dissolves oils and grease, enabling detergents to remove oily stains effectively. ✓Pu

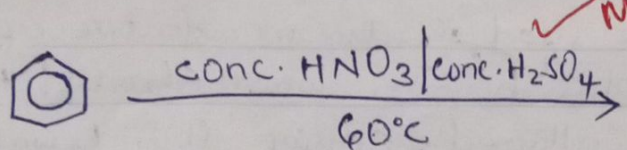
The intermediate is a liquid at room temperature; making it easy to pump, mix, and process during detergent manufacture ✓Pu

The intermediate has low solubility in water since it is non-polar; allows it to react effectively with sulphonating agents before being converted into the water-soluble detergent. ✓Pu

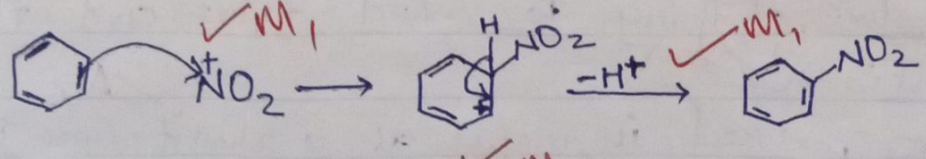
The intermediate contains reactive benzene ring which readily undergoes sulphonation to form linear alkylbenzene sulphonate. ✓Pu

2Pu = 02

Mechanism for the reaction involved in the formation of compound X which is  from benzene.

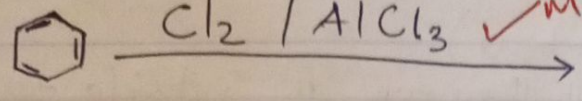


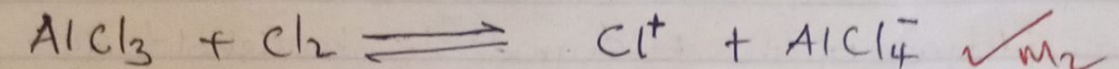
$HNO_3 + 2H_2SO_4 \rightleftharpoons \overset{+}{N}O_2 + H_3O^+ + 2HSO_4^-$ Rej. Unbalanced. ✓M1

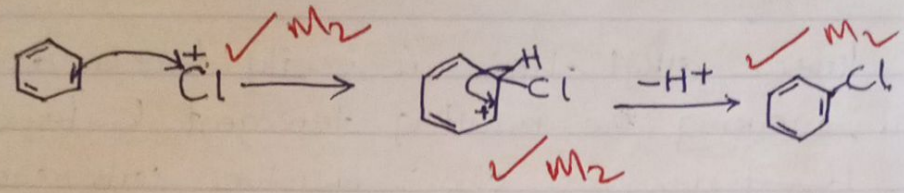


Mechanism involved in formation of compound 2,  from benzene

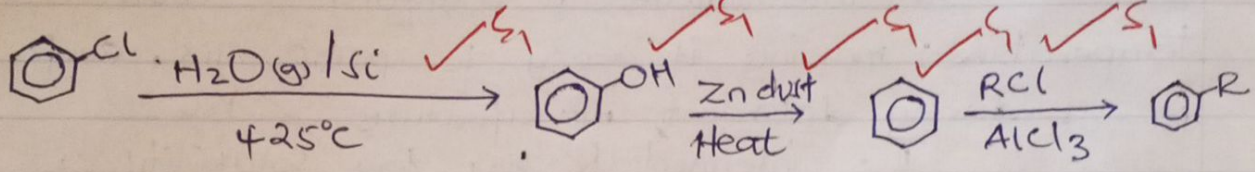
6M, +



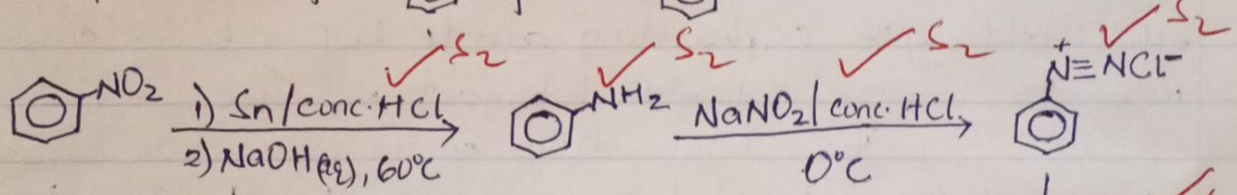




Synthesis of alkylbenzene,  from 



Synthesis of  from 



Accept any other correct route

6M, + 6M2 = 03
4-5M, + 4-5M2 = 02
<3M, + <3M2 = 00

5S1 + 5S2 = 03
3-4S1 + 3-4S2 = 02
Accept any other correct route
<3S1 + <3S2 = score 00

Phenol produced destroys bacteria and other microorganisms by denaturing their protein and disrupting cell membrance and are used in the manufacture of disinfectants and antiseptics however when absorbed through the skin, inhaled or swallowed, causes skin burns and damage of the nervous system mitigated by wearing gloves, protective clothing to avoid direct contact with the skin. AM Aa Ac

Benzene produced is useful as a solvent since it dissolves many non-polar substances such as oils, fats, etc Aia

however long term exposure damages the bone marrow, reducing the production of blood cells and increasing the risk of leukemia and other blood cancers ✓Ad mitigated by Wearing approprate personal protective equiptments e.g masks, protective clothings to limit exposure. ✓Am

2Aa + 2Ad + 2Am = 03

2Aa + 2Ad [2Am] = 02

2Aa + 2Ad

< 2Aa + < 2Ad + < 2Am = 00

Item 6

Suitability of the selected raw materials.

Propanone hydrazone is a polar compound since it contains C=N and $-NH_2$ groups; so it dissolves well in many polar organic solvents used in polymer additive. ✓Pu

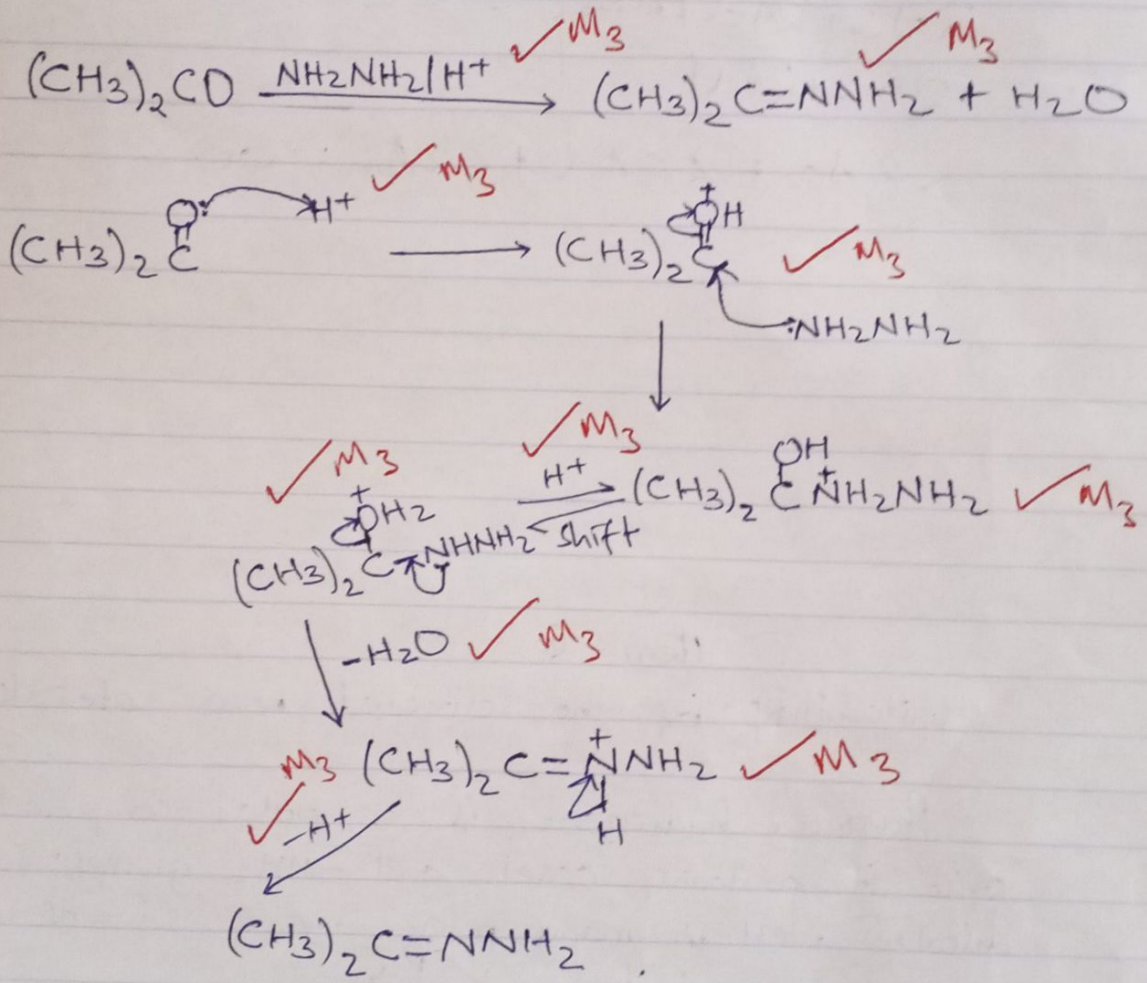
Propanone hydrazone has low boiling point due to strong dipole-dipole interractions between its molecules making it less volatile, reducing evaporating looses during processing. ✓Pu

Cyclohexene is non-polar; contains a hydrocarbon ring with a carbon-carbon double bond making it non-polar and compatible with many organic polymers used in coatings. ✓Pu

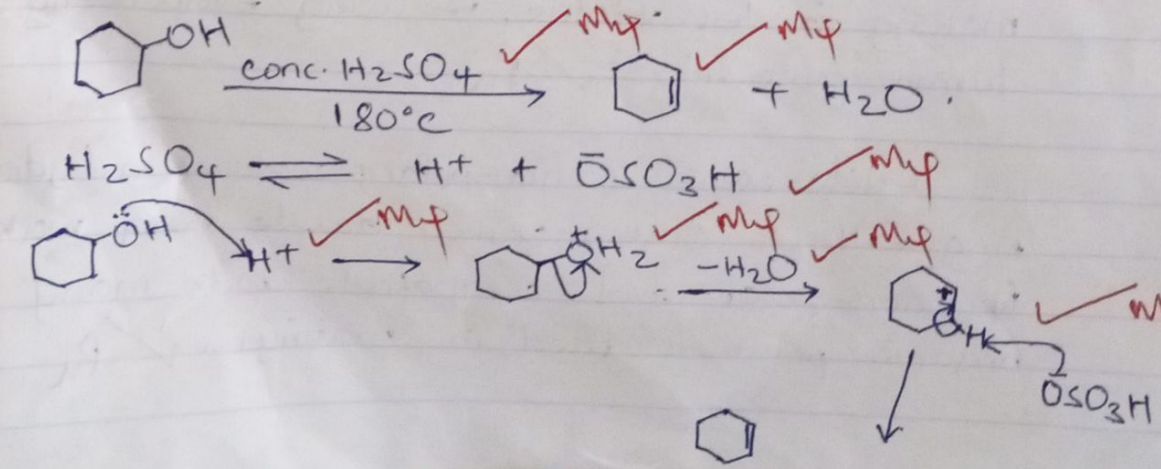
Cyclohexene is less soluble in water due to its hydrophobic nature; helps produce coatings with improved water resistance and durability. ✓PU

2Pu = 02

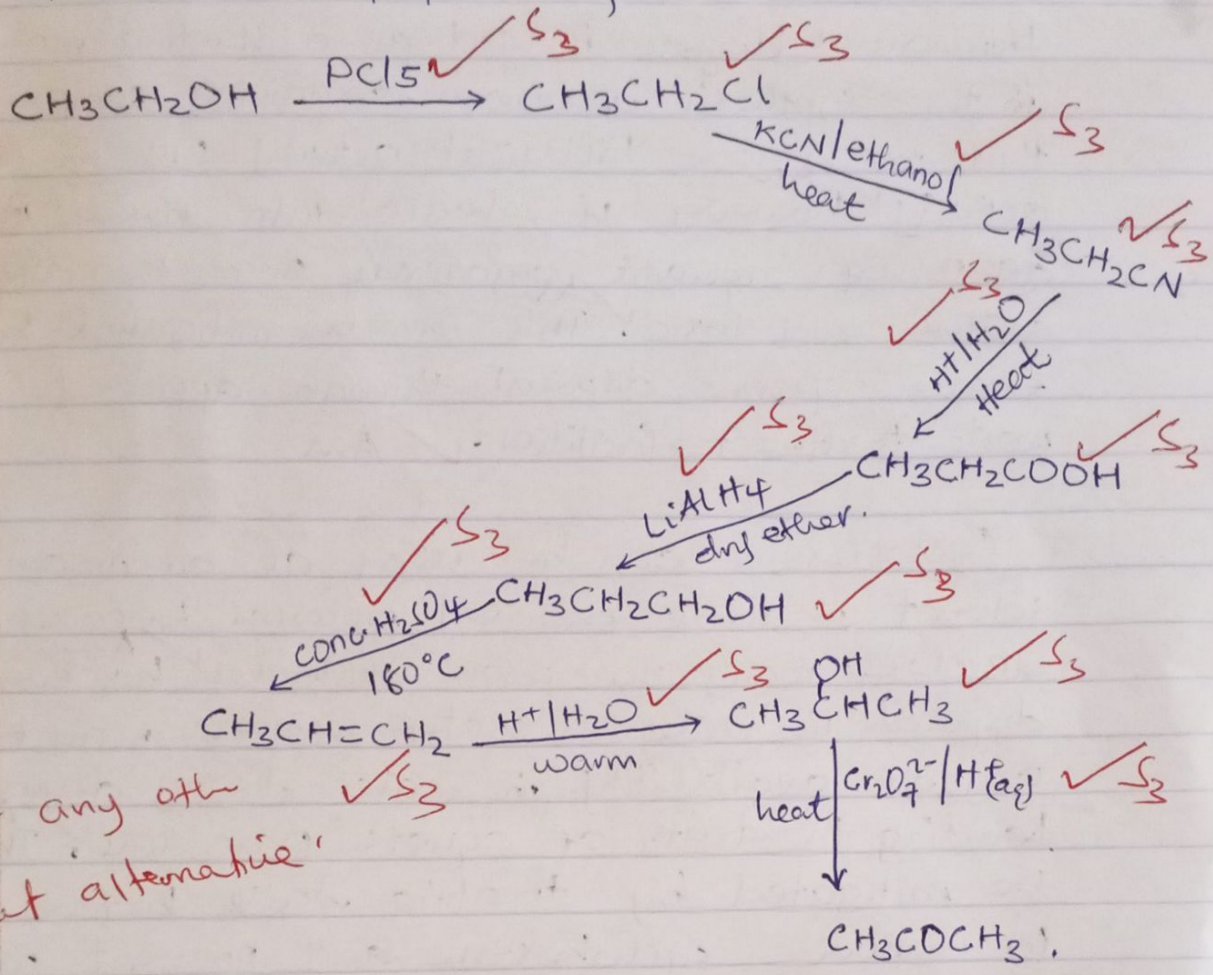
Mechanism for the reaction involved in the formation of propanone hydrazone from propanone.



Mechanism for the reaction involved in the formation of cyclohexene from cyclohexanol.

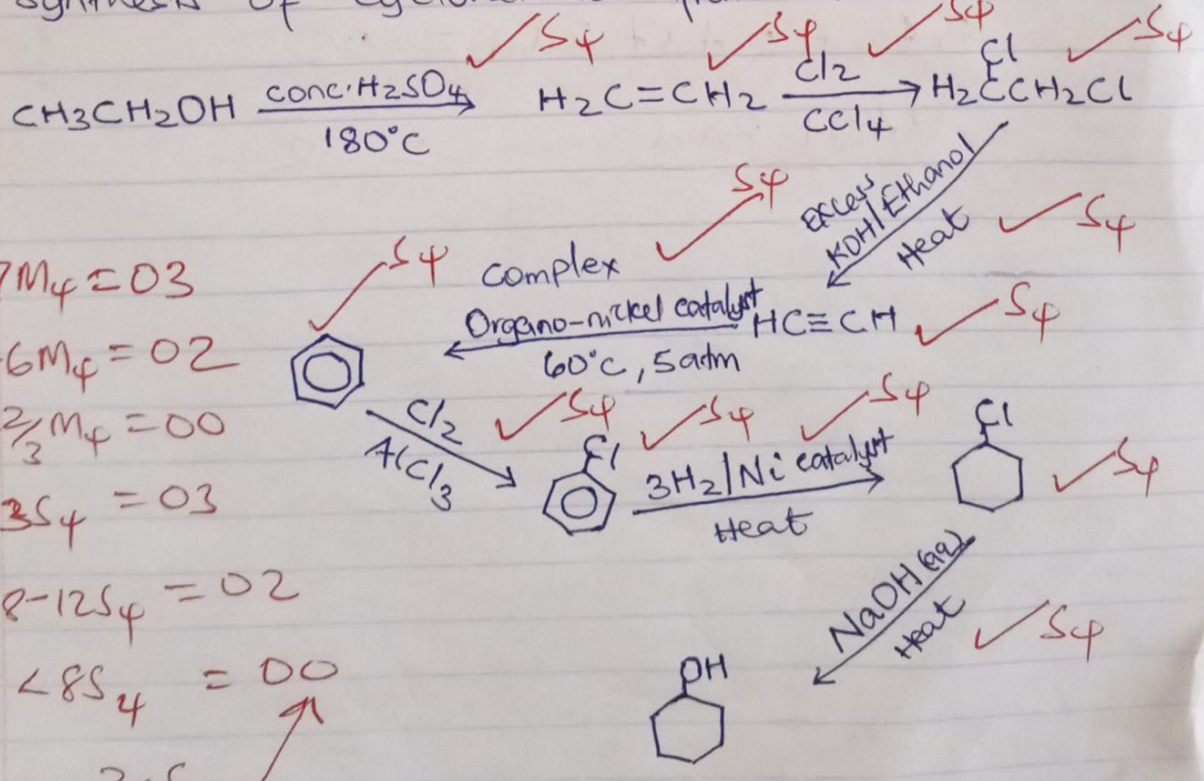


Synthesis of propanone from ethanol.



Accept any other correct alternative

Synthesis of cyclohexanol from ethanol



$10M_3 + 7M_4 = 03$

$7M_3 + 5\text{-}6M_4 = 02$

$<\frac{2}{3}M_3 + <\frac{2}{3}M_4 = 00$

$13S_3 + 13S_4 = 03$

$8\text{-}12S_3 + 8\text{-}12S_4 = 02$

$<8S_3 + <8S_4 = 00$

or

$<\frac{2}{3}S_3 + <\frac{2}{3}S_4$

Propanone hydrazone is used as a starting material in the synthesis of nitrogen-containing compounds that may have biological activity ✓Aa such as antimicrobial activity however if release into rivers, it causes death of aquatic organisms ✓Ad since they are biologically active and toxic. This can be mitigated by ensuring proper disposal through approved chemical waste treatment facilities. ✓Am

Cyclohexene can be used as an organic solvent in some chemical processes because it ✓Aa dissolves many non-polar substances however improper disposal in water in water bodies may reduce oxygen availability in water during degradation ✓Ad leading to death of aquatic organisms. This can be mitigated by treating waste before discharge to remove cyclohexene and prevent contamination of aquatic ecosystems. ✓Am

2Aa + 2Ad + 2Am = 03

2Aa + 2Ad = 02